

LIFEFORCE

Metabolic Plan



Overview

This Metabolic Plan provides a diet and lifestyle program focused on optimizing your metabolic health which includes how efficient your metabolism is converting food into energy. Nearly 93% of American adults are metabolically unhealthy, having one or more markers of metabolic dysfunction (adiposity, blood glucose, blood lipids, blood pressure, and cardiovascular disease). However, metabolic conditions are preventable and oftentimes reversible with diet and lifestyle modifications. A balanced diet rich in nutrient-dense foods, such as low-glycemic fruits, non-starchy vegetables, quality proteins, and healthy fats supports stable blood sugar levels and efficient energy use. Additionally, regular exercise, quality sleep, and stress management further contribute to a healthy metabolism. By prioritizing these factors, you can improve your metabolic health, reduce the risk of metabolic disorders, and promote long-term wellness. Change requires effort though so take advantage of your dedicated health coach to help motivate you and personalize this Plan to meet *your* dietary needs and lifestyle preferences.



Nutrition

Nutrition plays a pivotal role in influencing the risk and progression of metabolic conditions (e.g., insulin resistance, diabetes). The below nutrition plans focus on consuming a variety of low-glycemic fruits and non-starchy vegetables, quality proteins, and healthy fats, while avoiding trans fats, processed foods, added sugars, and limiting other high-carbohydrate foods to keep blood sugar and insulin levels within range and stable (i.e., avoiding frequent spikes). We suggest minimizing alcohol as well as it is a known toxin and recent studies are challenging previously touted health benefits of alcohol so “less is better” is the more cautionary approach.

Nutrition Plans

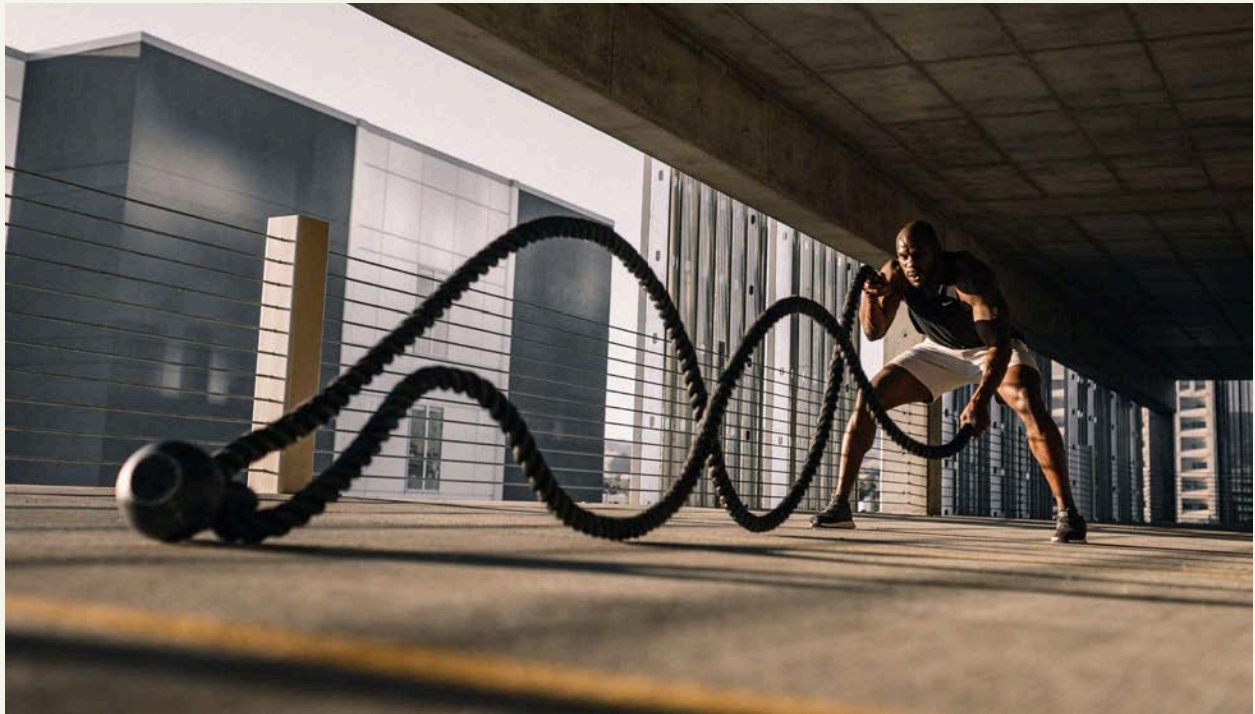
Whole-Foods Nutrition Plan–Lower-Carb Version. Consists of real, nutrient-dense foods, rich in colorful fruits and non-starchy vegetables, quality proteins and dairy, healthy fats, and nuts/seeds. This plan limits starchy foods such as grains, legumes, starchy vegetables, and avoids processed foods, refined carbohydrates, and added sugars.

Mediterranean Diet–Lower-Carb. Combines the principles of the traditional Mediterranean diet with a low-carbohydrate approach. Emphasizes the consumption of whole foods, high-quality fats (e.g., olive oil, nuts), plenty of non-starchy vegetables, low-glycemic fruit, moderate amounts of lean protein (e.g., fatty fish and poultry), and limited dairy. Limits or eliminates grains, legumes, starchy vegetables, refined carbohydrates, and added sugars.

Ketogenic Diet. "Keto" is a high-fat, low-carbohydrate diet which induces the body to burn primarily fat for fuel instead of carbohydrates. Includes quality meat, poultry, fish, eggs, full-fat dairy products, nuts and seeds, non-starchy vegetables, healthy fats and oils, and low-glycemic

fruits. This diet is restrictive on carbohydrates thereby limiting grains, legumes, starchy vegetables, and avoiding processed foods, refined carbohydrates, and added sugars.

Diets are most impactful when personalized which may include exploring other nutrition plans or specific food restrictions so work with your dedicated health coach in selecting an appropriate plan for you.



Physical Activity

Regular exercise is vital for maintaining and improving metabolic health. Physical activity enhances the body's ability to regulate blood sugar levels, improves insulin sensitivity, reduces inflammation, and helps maintain a healthy weight by increasing energy expenditure.

A complete exercise framework consists of four components: stability (foundation), strength/muscle mass, aerobic workout/Zone 2, and anaerobic/Zone 5 training, however if you're unsure where to begin or have limited time, prioritize aerobic/Zone 2 and strength training. Connect with your health coach to learn more!

General Framework

Strength Training: Strength training builds muscle mass, improves metabolic rate, helps stabilize and regulate blood sugar, and enhances insulin sensitivity. Aim for at least 2 sessions per week.

Aerobic/Zone 2 Training: Zone 2 training is moderate-intensity exercise at 60-70% of maximum heart rate. This type of exercise enhances the body's ability to burn fat for fuel, improves insulin sensitivity, and increases mitochondrial efficiency, leading to better energy production and utilization. Aim for 2-4 sessions per week, at least 45 minutes per session.

Anaerobic/Zone 5 Training: Zone 5 training is high-intensity exercise at 90-100% of maximum heart rate. This intense level of training enhances anaerobic capacity, boosts metabolic rate, improves insulin sensitivity, contributing to better blood sugar regulation and overall metabolic function. Aim for 1-2 sessions per week.

Stability Training: Stability training includes exercises that improve core strength, help with coordination and balance, improve proprioception, and build a foundation to prevent injury. Aim for a few minutes of stability training per day.



Sleep

Adequate sleep is essential for maintaining optimal metabolic health. Quality sleep regulates crucial hormones like insulin, cortisol, and leptin, which play significant roles in appetite control, glucose metabolism, and energy balance. Chronic sleep deprivation can disrupt these hormonal balances, leading to increased insulin resistance, higher blood sugar levels, and a greater risk of obesity and type 2 diabetes. Moreover, poor sleep quality can raise stress levels and promote inflammation, further compromising metabolic function. Prioritizing good sleep hygiene is vital for supporting metabolic health and overall well-being.

Sleep Hygiene

These are general recommendations, but please reach out to your coach for a more personalized approach.

Get 7-9 hours of sleep. The optimal quantity of sleep varies per individual. The goal is to wake up feeling rested.

Maintain a consistent bedtime. Sleep consistency (going to bed around the same time each night) may be more important for overall longevity than the amount of time spent asleep. Additionally, your bedtime largely depends on your chronotype.

Sync circadian rhythms. Get safe sun exposure during the day and minimize exposure to bright, green/blue light in the evening to balance circadian rhythms. This is extremely important to protect melatonin (a powerful antioxidant) production as well.

Keep the bedroom dark, cool, and quiet. Use blackout blinds or an eye mask. Optimal sleeping temperature is 62-68 degrees. If noises are disruptive, wear ear plugs or play soothing background sounds.

Wind down. Add a calming wind-down routine before lights out. Take a hot bath to facilitate dropping your core body temperature which is important to initiate and maintain quality sleep. This happens because of vasodilation (widening of blood vessels) to release that pent up heat. Also consider deep breathing, meditation, journaling, reading, or listening to soothing music.



Detoxification

Environmental toxins, such as heavy metals, pesticides, and industrial chemicals, pose significant risks to metabolic health by disrupting endocrine function, leading to hormonal imbalances that affect appetite regulation, insulin sensitivity, and fat storage. Chronic exposure to these toxins can increase the risk of obesity, type 2 diabetes, and other metabolic disorders by promoting inflammation, oxidative stress, and cellular dysfunction. Reducing exposure to these toxins or by improving detox pathways through diet and lifestyle modifications can help mitigate the risk of metabolic diseases.

Tips for Detoxification

Quit smoking, vaping, and chewing tobacco. Smoking has been identified as a significant risk factor for the development of metabolic diseases through the harmful effects of oxidative stress, inflammation, and mitochondrial damage.

Reduce exposure to obesogens. Obesogens are environmental chemicals (e.g., BPA, phthalates, PFOA, organotins, and certain pesticides) that can disrupt the body's normal metabolic processes and contribute to obesity and related metabolic disorders. They are found in various sources like plastics, pesticides, and processed foods, and promote the development of fat cells, alter metabolism, and increase insulin resistance, leading to weight gain and increased risk of metabolic conditions. Reduce exposure by buying organic, avoiding plastics, using clean products at home, and drinking filtered water.

Consume detoxifying foods. Certain foods are known for their detoxifying properties including cilantro, garlic, wild blueberries, lemon water, spirulina (microcystin-free), chlorella, cruciferous vegetables, broccoli sprouts, bitter greens, turmeric, ginger, beets, and citrus fruits. Detoxifying herbs include basil, mint, rosemary, and parsley. Detoxifying teas include fennel, dandelion, turmeric, milk thistle, ginger, lemon, and green tea.

Stay hydrated, but also sweat! Adequate hydration helps the kidneys and liver efficiently flush out toxins from the body. Aim for at least half of your body weight in ounces per day of filtered water. Sweat often to eliminate some toxins through the skin, supporting overall detoxification.



Stress Management

Managing stress is critical for metabolic health as chronic stress causes the body to release cortisol, our main stress hormone. Cortisol then tells the liver to create sugar and dump it into the bloodstream for the body to use as energy (to escape the proverbial tiger) which puts your blood sugar on a rollercoaster ride, leading to inflammation. Chronic stress can also disrupt sleep and increase appetite, particularly for high-calorie and sugary “comfort” foods, contributing to weight gain and obesity. By addressing stress effectively though, you can reduce the risk of developing metabolic disease.

Tips to Manage Stress

Incorporate breathwork techniques. Deep breathing exercises, such as diaphragmatic breathing or box breathing, can activate the body's relaxation response and reduce stress levels.

Try meditation. Meditation involves training your mind to focus and redirecting thoughts back to the practice which can lead to a state of heightened awareness, mental clarity, and inner calm. Regular meditation has been shown to activate the body's relaxation response and lower levels of stress hormones.

Connect with nature. Spending time in nature has a profound impact on stress reduction. Nature has a calming effect on the nervous system, lowers cortisol levels, and promotes a sense of tranquility and well-being. In fact, if you take a walk or hike in nature, you're checking the box on physical activity as well!

Do things that you enjoy. One of the very best ways to manage your stress is simply by doing things that fill your cup. If you love watching a movie with a loved one, do that! If it's playing board games with friends, do that. If it is shopping, exercising, or cooking, it doesn't matter; if you like it, then do it!



Community & Connectedness

Loneliness can significantly impact metabolic health by contributing to chronic stress, which elevates cortisol levels and can lead to insulin resistance, increased blood sugar levels, and weight gain. The lack of social interaction and support often associated with loneliness can exacerbate unhealthy behaviors such as poor diet, lack of physical activity, and disrupted sleep patterns. These combined factors increase the risk of developing metabolic disorders. Addressing loneliness through social engagement, supportive relationships, and mental health interventions can help mitigate these negative effects and promote better metabolic health.

Conclusion

Metabolic health is paramount for overall wellness, affecting energy production, weight management, and disease prevention. A balanced approach combining regular exercise, proper nutrition, quality sleep, stress management, efficient detoxification, and strong social connections reduces inflammation and optimizes metabolism. Change takes effort so work with your health coach to help implement these recommendations. Since metabolic disorders are highly preventable, applying these changes can set you on a path to a vibrant and healthy life.